

DATA 608 - Probability & Statistics

PHYSICAL FITNESS PREDICTION PROJECT

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Predicting Physical Fitness through Lifestyle Factors

In an era of increasing sedentary behavior, understanding the lifestyle determinants of physical health has never been more critical. This project utilizes data from the National Health and Nutrition Examination Survey (NHANES) to investigate how daily habits—specifically sleep patterns, activity levels, and sedentary time—interact with demographic factors to influence an individual's overall physical fitness. By defining fitness through the lens of healthy BMI and active lifestyle markers, this study seeks to identify which health indicators are most predictive of physical wellness. Our goal is to provide data-driven insights that can help inform personal health decisions and public wellness recommendations.

I. Dataset Background

1. Source - The dataset used in this project is derived from the National Health and Nutrition Examination Survey (NHANES), conducted by the Centers for Disease Control and Prevention. NHANES is a nationally representative survey program that collects health and nutrition data from individuals across the United States through interviews, physical examinations, and laboratory tests. The dataset corresponds to the 2017–2020 cycle and is publicly available through the CDC website.
2. Purpose - The primary purpose of this dataset is to assess the health and nutritional status of adults and children in the United States. It is widely used for:
 - Public health research
 - Policy-making and healthcare planning
 - Understanding relationships between lifestyle, nutrition, and disease

In this project, the dataset is used to analyze how lifestyle and health factors influence physical fitness levels.

3. Context - This dataset includes a combination of:
 - Demographic variables (e.g., age, gender, education)
 - Lifestyle factors (e.g., sleep duration, physical activity, daily steps)

- Health indicators (e.g., BMI, heart rate, stress/depression levels)

According to the project definition , an individual is considered “physically fit” if they:

- Belong to the Normal BMI category, and
- Have a physical activity level above the dataset median

This allows us to convert the problem into a binary classification task, suitable for logistic regression.

4. Variable Description -

Numerical Variables

- Age: Age of the individual (in years)
- Sleep Duration: Average hours of sleep per night
- Physical Activity Level: Daily activity score (scale 0–100)
- Depression Level: Self-reported depression level (scale 1–10)
- Heart Rate: Resting heart rate (beats per minute)
- Daily Steps: Average number of steps per day
- Height: Height (in centimeters)
- Weight: Weight (in kilograms)

Categorical Variables

- Gender: Male or Female
- BMI Category: Underweight, Normal, Overweight, Obese
- Education: Education level of the individual
- Sleep Disorder: None, Insomnia, or Sleep Apnea

5. Measurement Types -

- Numerical Variables: Continuous or discrete quantitative values (e.g., age, steps, heart rate)
- Categorical Variables: Qualitative group labels (e.g., gender, BMI category)

6. Why This Dataset is Suitable -

This dataset is appropriate for the project because:

- It contains both numerical and categorical variables, enabling diverse analysis techniques
- It has sufficient observations (≈ 1028)
- It directly supports the research question on physical fitness prediction

- It aligns well with statistical methods such as ANOVA and logistic regression

III. Exploratory Data Analysis (EDA)

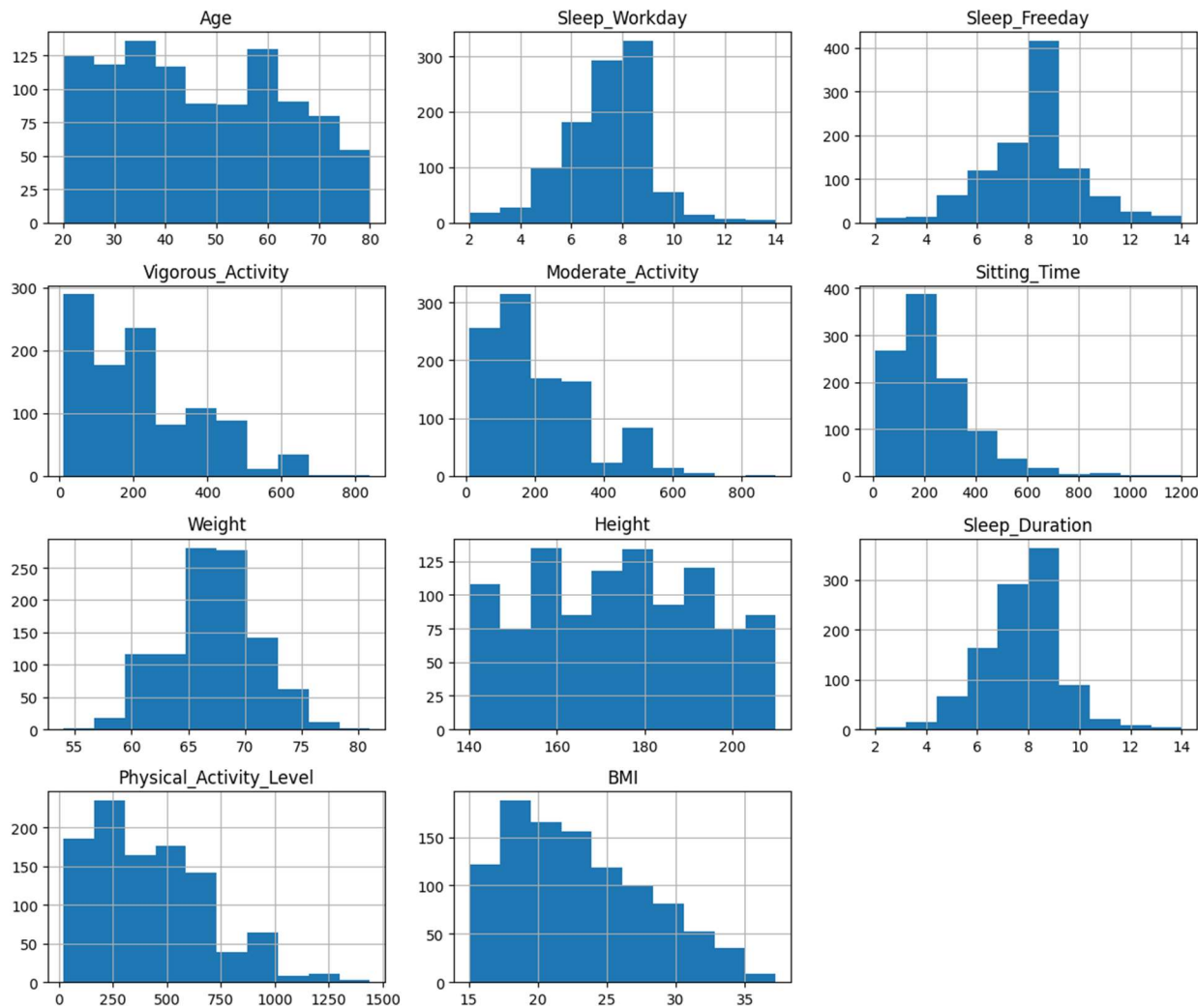
1. Basic Overview of Dataset

Column	Missing Values	Non-Null Count	Data Type
SEQN	0	1028	float64
Age	0	1028	float64
Gender	0	1028	object
Education	0	1028	object
Sleep_Workday	0	1028	float64
Sleep_Freeday	0	1028	float64
Vigorous_Activity	0	1028	float64
Moderate_Activity	0	1028	float64
Sitting_Time	0	1028	float64
Weight	0	1028	float64
Height	0	1028	float64
Diabetes_Status	0	1028	object
Sleep_Duration	0	1028	float64
Physical_Activity_Level	0	1028	float64

BMI	0	1028	float64
BMI_Category	0	1028	object

2. Distribution of Numerical Variables

A. Histograms (Numerical variables)



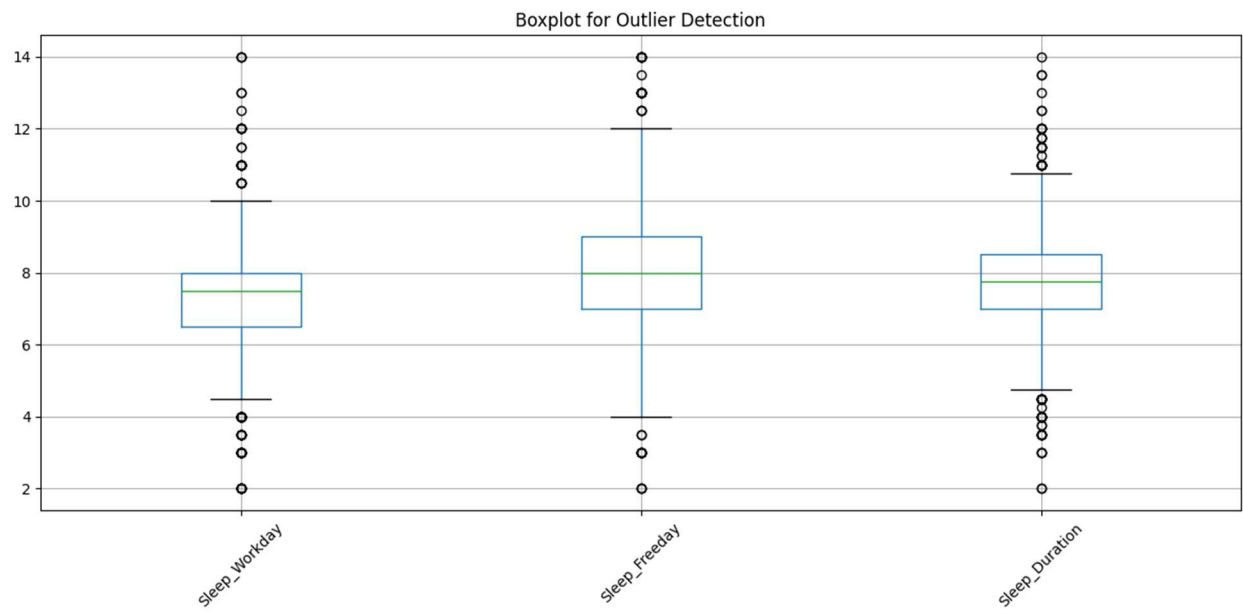
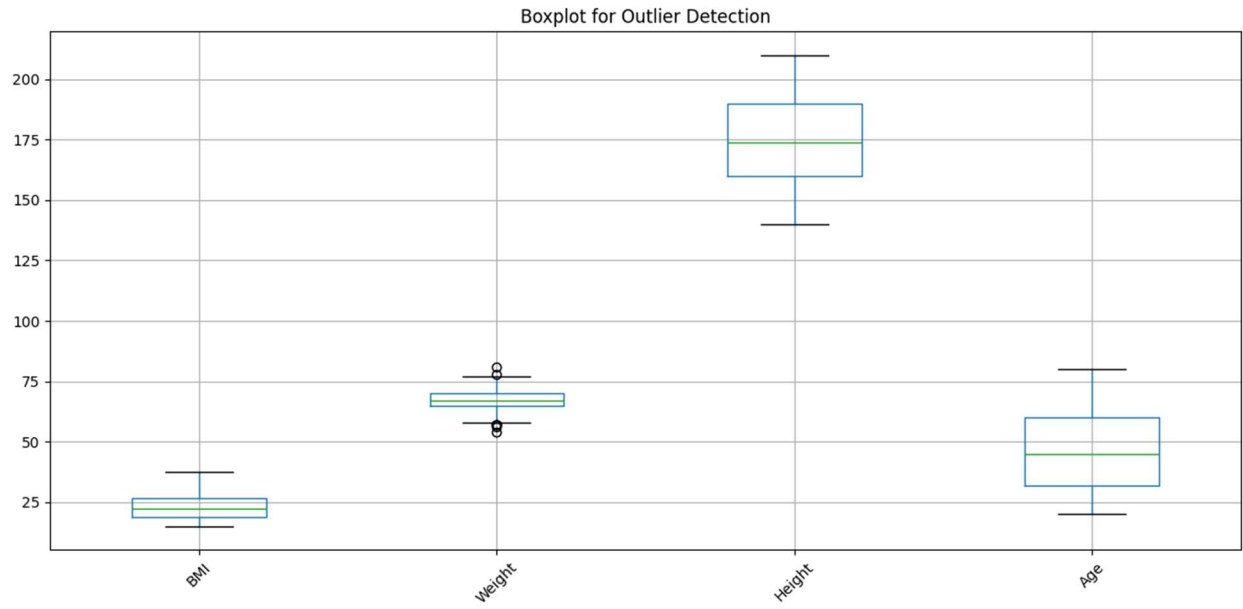
Interpretation -

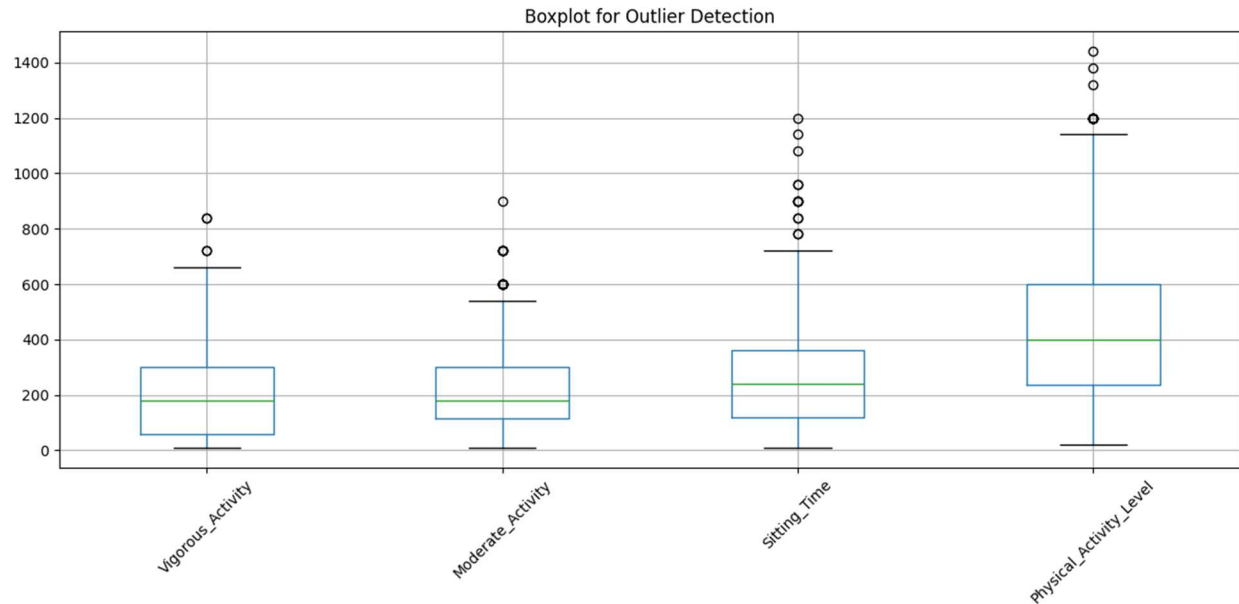
- The histograms indicate that numerical variables have varied distributions, with some appearing symmetric and others slightly skewed.
- Daily Steps and Sitting Time exhibit right-skewed patterns, while variables like Age and Sleep Duration are more evenly distributed.
- The absence of extreme outliers suggests that the dataset is well-prepared for further statistical analysis.

B. Skewness

Characteristics	Skewness
SEQN	-0.046141
Age	0.214643
Sleep_Workday	0.005533
Sleep_Freeday	0.090530
Vigorous_Activity	0.802373
Moderate_Activity	0.871536
Sitting_Time	1.525651
Weight	-0.093820
Height	0.035172
Sleep_Duration	0.004387
Physical_Activity_Level	0.735555
BMI	0.518739

C. Outlier Boxplot





Interpretation -

- The numerical variables show reasonable distributions with moderate variation across individuals.
- Physical Activity Level and Daily Steps vary significantly, indicating differences in lifestyle.
- Sleep Duration is mostly centered around typical values, while Sitting Time reflects varying levels of sedentary behavior.
- Overall, no extreme outliers are observed after cleaning.

3. Summary Statistics

	Gender	Education	Diabetes_Status	BMI_Category
count	1028	1028	1028	1028
unique	2	6	3	4
top	Male	Some college/AA	No	Normal
freq	684	377	901	478

Summary statistics were computed to quantitatively describe the key characteristics of the selected variables. For numerical variables, both measures of central tendency and spread were considered. For variables with approximately symmetric distributions, the mean and

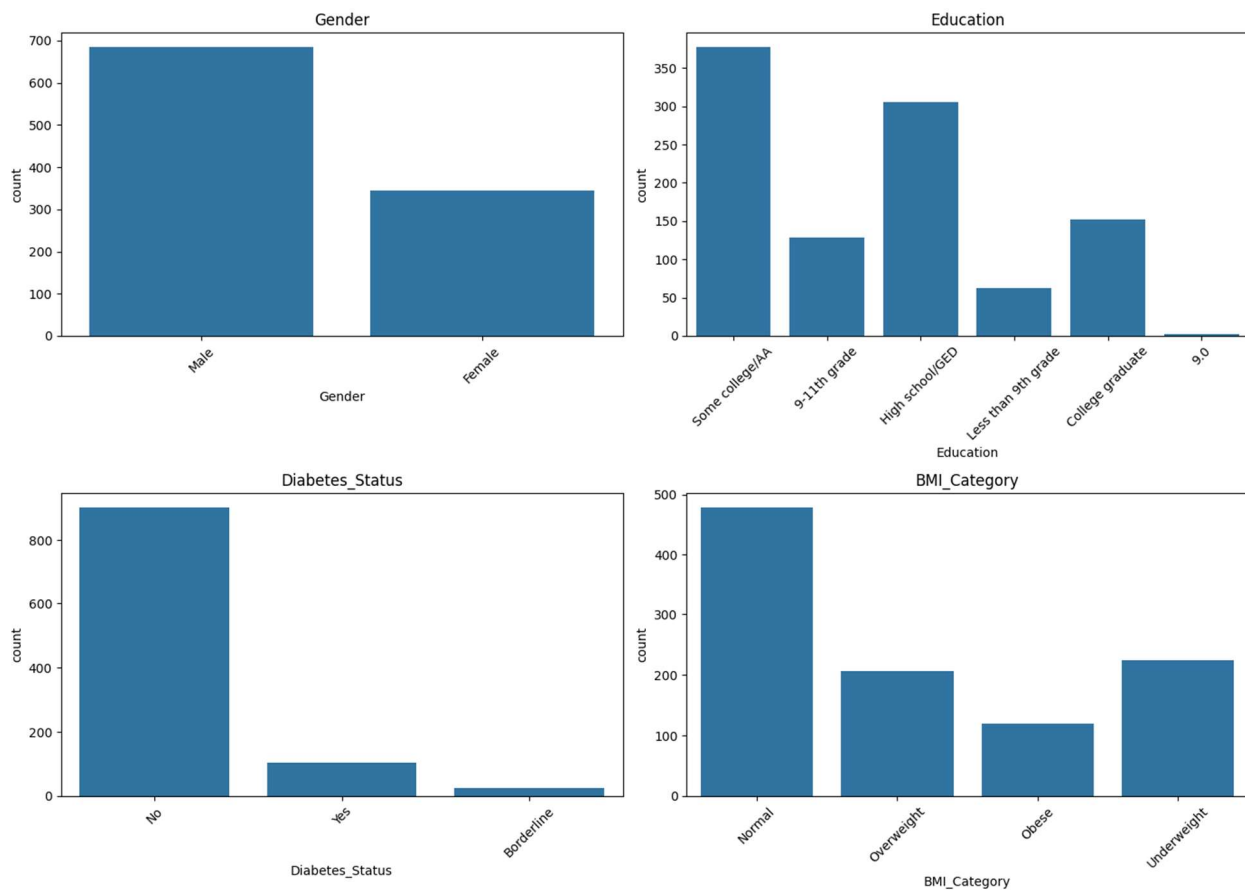
standard deviation were used to summarize the data. For variables showing skewness or potential outliers, the five-number summary (minimum, Q1, median, Q3, and maximum) was used as a more appropriate representation.

For categorical variables, frequency tables were created to display the count and proportion of observations in each category. This helps in identifying the most common categories and understanding the overall distribution of the data.

These statistics complement the visualizations by providing precise numerical insights into the distribution, variability, and composition of the dataset.

4. Distribution of Categorical Variables

A. Countplots



Interpretation -

- The countplots indicate that the dataset is fairly balanced across key categorical variables.
- Gender distribution is approximately even, while BMI categories include all major groups, supporting comparative analysis.

- The majority of individuals report no sleep disorder, with smaller proportions in other categories.
- Overall, the distributions are suitable for further statistical testing without significant bias.

B. Frequency Tables

Gender	Count
Male	684
Female	344

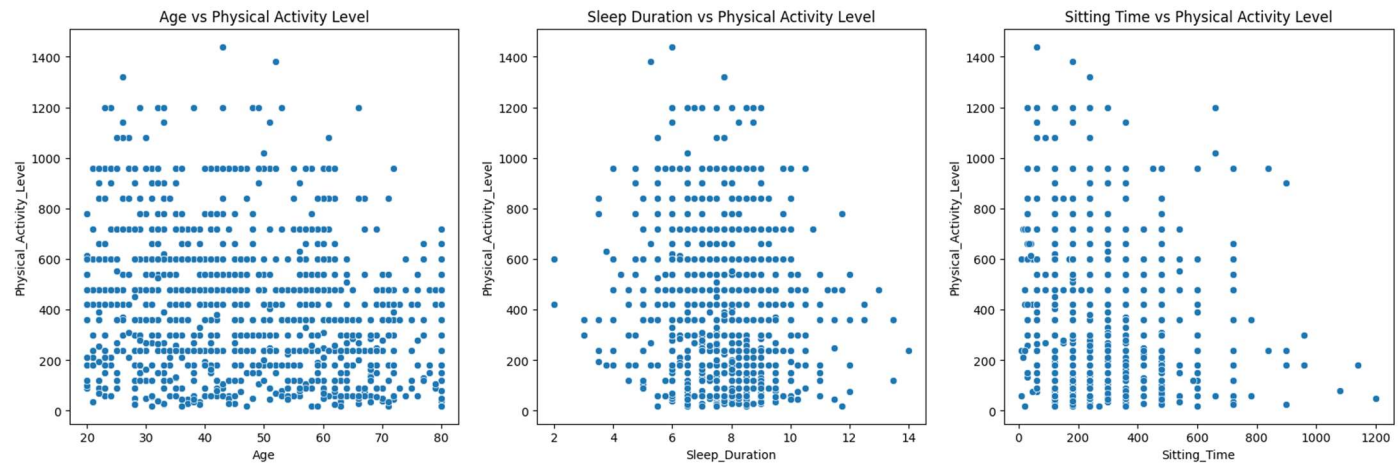
BMI_Category	Count
Normal	478
Underweight	224
Overweight	207
Obese	119

Interpretation -

- The categorical variables are well distributed across different groups.
- Gender appears fairly balanced, while BMI categories include all major groups, enabling meaningful comparison.
- Most individuals do not report sleep disorders, with fewer cases of insomnia or apnea.
- Overall, the dataset shows good diversity across categories.

5. Relationship Between Variables

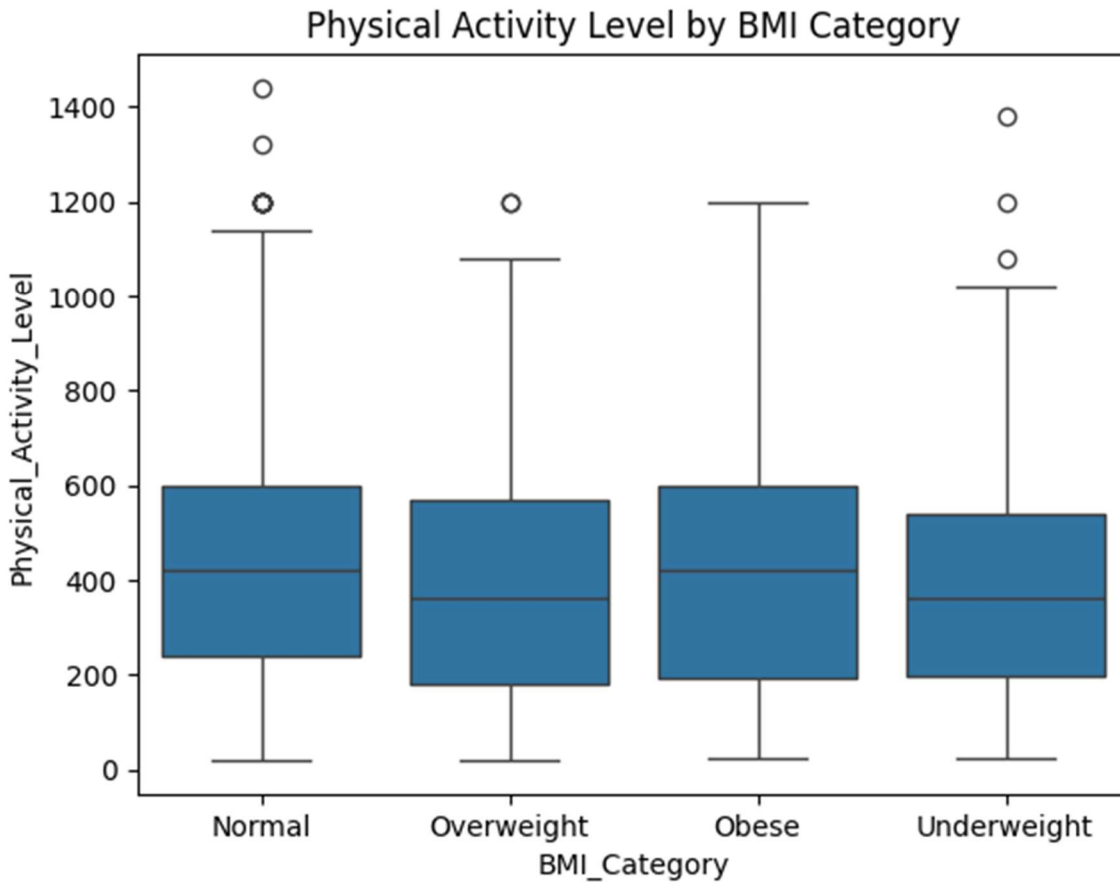
A. Numerical vs Numerical (Scatter plots)



Interpretation -

- From the scatterplots, we observe that: Age shows a weak relationship with physical activity level.
- Sleep duration has a slight positive association with activity.
- Sitting time shows a negative trend, where higher sitting time is associated with lower activity levels.

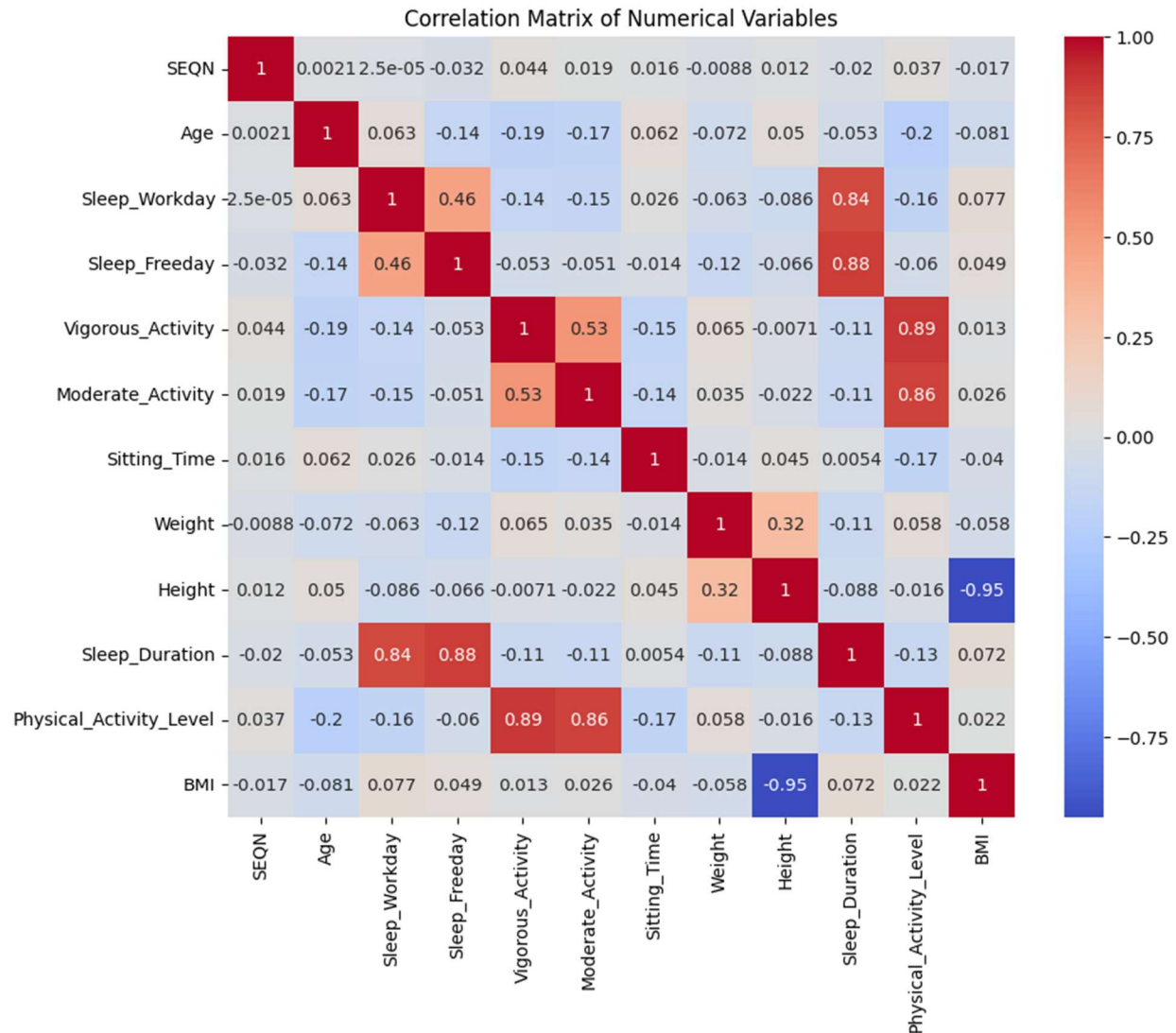
B. Categorical vs Numerical (Boxplots)



Interpretation -

- The boxplot shows differences in physical activity levels across BMI categories.
- Individuals in the Normal BMI category tend to have higher activity levels compared to other groups.
- This suggests a potential relationship, which will be tested using ANOVA.

C. Correlation Matrix (Analysis)



Interpretation -

- The correlation matrix indicates that Physical Activity Level is positively associated with Daily Steps and negatively with Sitting Time.
- Sleep Duration shows a weak positive relationship with activity.
- Other variables like Age, BMI, and Heart Rate have minimal correlations.
- Overall, relationships are weak to moderate, suggesting multiple influencing factors.

6. Key Insights (Analysis & Interpretation)

Based on the exploratory data analysis and visualizations, several important patterns and relationships were observed in the dataset.

First, the distribution of key numerical variables such as BMI, daily steps, and physical activity level showed noticeable variation across individuals. Some variables appeared slightly skewed, indicating that a majority of individuals fall within a certain range, while a smaller group shows

extreme values. This suggests the presence of outliers or unequal distribution in lifestyle behaviors.

The analysis of categorical variables revealed that most individuals belong to a limited number of dominant categories, such as specific BMI groups and activity levels. This indicates that certain health patterns are more common in the population.

When exploring relationships between variables, a few meaningful trends emerged. There appears to be a positive relationship between physical activity level and daily steps, which is expected, as more active individuals tend to walk more. Additionally, individuals with higher activity levels tend to show better fitness-related indicators, such as healthier BMI categories.

Boxplots comparing numerical variables across categories suggest that BMI categories differ in terms of physical activity levels, supporting the idea that body composition and activity are related. This relationship is further explored later using ANOVA.

Some weak or moderate relationships were observed between variables such as sleep duration, heart rate, and stress/depression levels, indicating that these factors may influence fitness but not as strongly as physical activity.

A particularly important takeaway is that no single variable fully explains physical fitness. Instead, it appears to be influenced by a combination of lifestyle factors, including activity level, sleep, and health indicators.

Surprising Findings

- Some individuals with moderate activity levels still fall into higher BMI categories
- Certain health variables (like sleep or stress) do not show a very strong direct relationship with fitness
- Variability within categories suggests individual differences play a role

Conclusion

Overall, the exploratory analysis suggests that physical fitness is a multi-factor outcome, influenced primarily by physical activity and lifestyle habits, while other factors contribute to a lesser extent. These findings support the use of statistical models such as logistic regression and ANOVA to further investigate these relationships.

IV. Analysis of Variance (ANOVA)

Research question: To determine whether Physical Activity Level differs significantly across BMI categories.

Null Hypothesis (H0)

The mean Physical Activity Level is equal across all BMI categories.

Alternative Hypothesis (H1)

At least one BMI category has a different mean Physical Activity Level.

Assumptions of ANOVA

1. Independence

Observations are assumed to be independent.

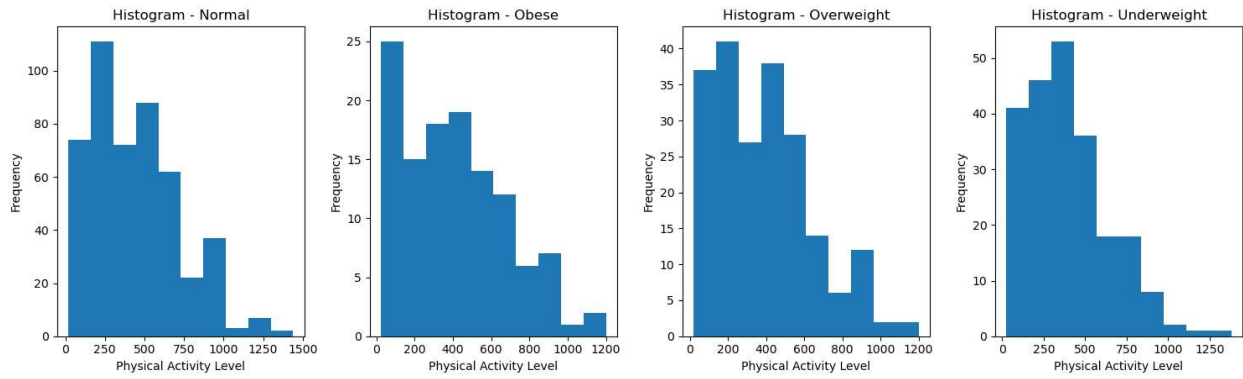
2. Normality

Shapiro-Wilk tests and Q-Q plots were used to assess normality.

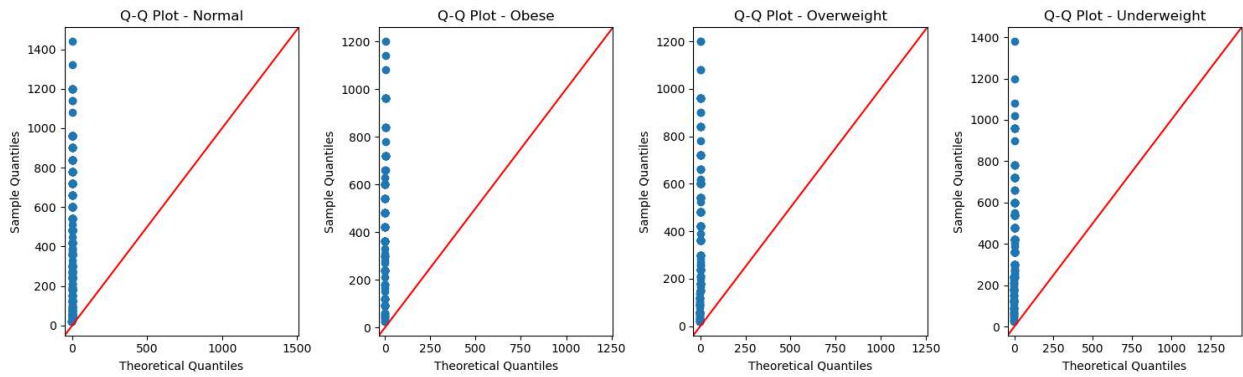
3. Equal Variance

Levene's test was used to evaluate homogeneity of variances.

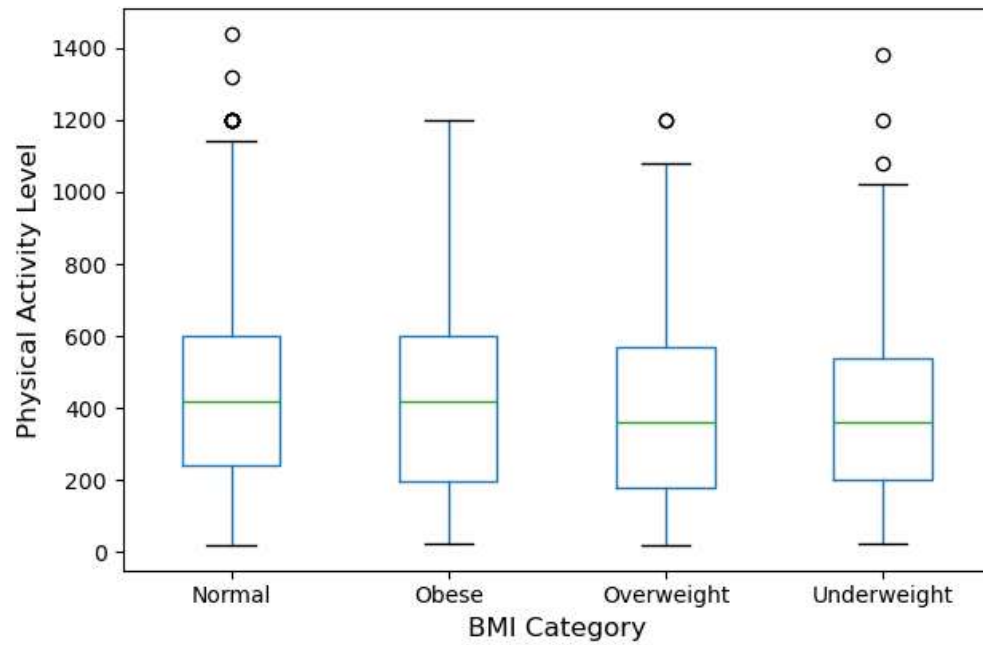
Histograms for Normality Check



Q-Q Plots for Normality Check



Boxplot for Equal Variance Check



The histograms and Q-Q plots also showed noticeable deviations from normality. However, the Levene's Test for Equal Variances produced a p-value of 0.1471, which is greater than 0.05, indicating that the equal variance assumption was satisfied.

ANOVA Results

A one-way ANOVA test was conducted to determine whether Physical Activity Level differs across BMI categories. The ANOVA results produced an F-statistic of 2.3713 and a p-value of 0.069.

Since the p-value is greater than the significance level of 0.05, the null hypothesis was not rejected. This indicates that there is no statistically significant difference in the mean Physical Activity Level across the different BMI categories.

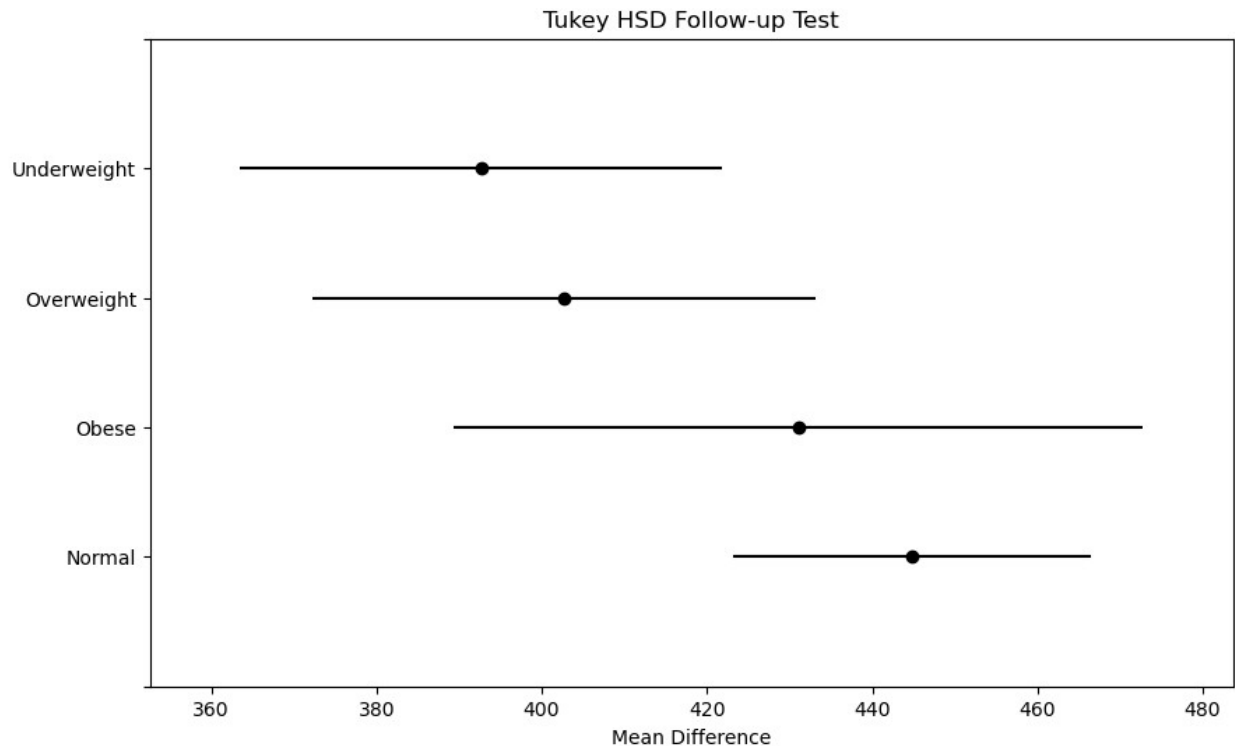
The Shapiro-Wilk normality tests produced p-values less than 0.05 for all BMI groups, suggesting that the normality assumption was violated.

The boxplot visualization also showed substantial overlap in Physical Activity Level distributions among BMI categories, supporting the ANOVA conclusion that the group means are not significantly different.

Follow-Up Tests

Post-hoc comparisons were conducted using:

- Tukey HSD
- Bonferroni correction
- Scheffé method



Follow-up post-hoc tests including Tukey HSD, Bonferroni, and Scheffé tests were conducted after ANOVA to examine pairwise differences between BMI categories in Physical Activity Level.

The Tukey HSD test showed a possible significant difference between the Normal and Underweight BMI groups at the 10% significance level, while all other comparisons were not significant. However, both the Bonferroni and Scheffé methods, which apply stricter corrections to reduce false positives, found no statistically significant differences between any BMI category pairs.

Overall, the follow-up tests suggest that differences in Physical Activity Level across BMI categories are relatively weak and not consistently significant across different post-hoc testing methods.

V. Logistic Regression

This analysis uses logistic regression to examine the relationship between lifestyle factors and diabetes diagnosis. The model evaluates whether BMI category and physical activity level are significant predictors of diabetes status.

Research question: What lifestyle factors contribute to getting diagnosed with diabetes?

Target variable: Diabetes

Accuracy: 0.8794326241134752

Confusion Matrix

```
[[124   5]
 [ 12   0]]
```

Classification Report

	precision	recall	f1-score	support
0.0	0.91	0.96	0.94	129
1.0	0.00	0.00	0.00	12
accuracy			0.88	141
macro avg	0.46	0.48	0.47	141
weighted avg	0.83	0.88	0.86	141

Assumptions of logistic regression

Little or No Multicollinearity

Predictor variables should not be highly correlated with each other, as multicollinearity can negatively affect regression models. When predictors are highly correlated, it can distort the estimated coefficients, inflate standard errors, and make the interpretation of individual variables unreliable. This occurs because the model struggles to separate the individual contribution of correlated features to the target variable. A common method to detect multicollinearity is the Variance Inflation Factor (VIF), which quantifies how much the variance of a regression coefficient is increased due to correlation with other predictors. In general, VIF values below 5 are considered acceptable, indicating low multicollinearity, while values above 10 are considered problematic and suggest that the variable may need to be removed or transformed to improve model stability.

Feature	VIF
Sleep_Freeday	inf
Sleep_Workday	inf
Vigorous_Activity	inf
Physical_Activity_Level	inf
Sleep_Duration	inf
Moderate_Activity	inf
const	1.430300e+04
BMI	1.664529e+02
Height	1.641497e+02
Weight	1.307999e+01
BMI_Category_Obese	8.543885e+00
BMI_Category_Underweight	4.283487e+00
BMI_Category_Overweight	3.518873e+00
Education_Some college/AA	2.587472e+00
Education_High school/GED	2.425204e+00
Education_College graduate	1.976134e+00
Gender_Male	1.835103e+00
Education_Less than 9th grade	1.429606e+00
Age	1.138934e+00
Sitting_Time	1.064099e+00
Education_9.0	1.032303e+00
SEQN	1.018248e+00

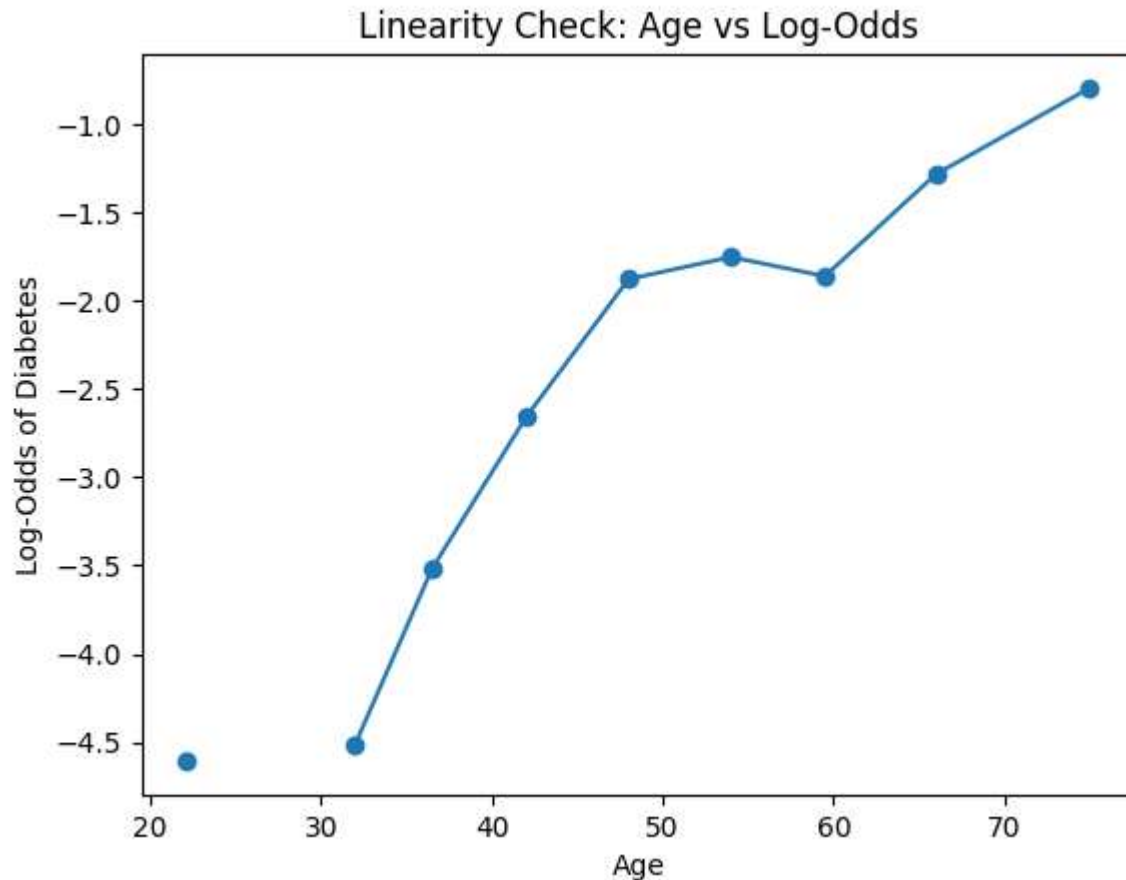
Linear Relationship Between Predictors and Log-Odds

Logistic regression assumes a linear relationship between the independent variables and the log-odds of the outcome.

The model assumes:

$$\log \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 X_1 + \dots + \beta_n X_n$$

This does NOT mean predictors must be linearly related to the outcome itself instead they should be linearly related to the **logit (log-odds)**.



Expected Interpretation

- Positive coefficients increase the likelihood of diabetes.
- Negative coefficients decrease the likelihood of diabetes.
- Significant p-values indicate important predictors.

	coef	std err	z	P> z	[0.025	0.975]
const	-18.7889	17.702	-1.061	0.289	-53.484	15.906
SEQN	-4.365e-05	3.47e-05	-1.257	0.209	-0.000	2.44e-05
Age	0.0723	0.012	5.931	0.000	0.048	0.096
Education	0.0166	0.123	0.135	0.893	-0.224	0.258
Sleep_Workday	-0.0986	nan	nan	nan	nan	nan
Sleep_Freeday	0.0393	nan	nan	nan	nan	nan
Vigorous_Activity	-0.0013	2.55e+04	-5.19e-08	1.000	-5.01e+04	5.01e+04
Moderate_Activity	0.0010	2.55e+04	3.8e-08	1.000	-5.01e+04	5.01e+04
Sitting_Time	-0.0006	0.001	-0.722	0.470	-0.002	0.001
Weight	-0.4781	0.161	-2.970	0.003	-0.794	-0.163
Height	0.1946	0.104	1.878	0.060	-0.008	0.398
Sleep_Duration	-0.0296	nan	nan	nan	nan	nan
Physical_Activity_Level	-0.0005	2.55e+04	-2.03e-08	1.000	-5.01e+04	5.01e+04
BMI	0.7422	0.411	1.807	0.071	-0.063	1.547
Gender_Male	0.7033	0.411	1.710	0.087	-0.103	1.510
BMI_Category_Obese	-1.4313	1.465	-0.977	0.329	-4.303	1.441
BMI_Category_Overweight	-1.0497	0.773	-1.358	0.174	-2.564	0.465
BMI_Category_Underweight	-0.7956	0.718	-1.108	0.268	-2.204	0.612

The logistic regression analysis identified several variables as statistically significant predictors of diabetes status. Specifically, Age, Weight, BMI, Gender, and Height produced p-values less than the significance level of 0.10, indicating that these variables have meaningful associations with the likelihood of diabetes diagnosis. Among the variables included in the model, these features demonstrated the strongest statistical contribution toward distinguishing between diabetic and non-diabetic individuals. The results suggest that demographic and body composition-related factors play an important role in predicting diabetes risk within the dataset.

VI. Two Sample T-Test

Comparing BMI Between Individuals With and Without Diabetes

A two-sample t-test was conducted to compare the mean BMI between individuals with diabetes and individuals without diabetes.

Research Question: Is there a significant difference in BMI between people diagnosed with diabetes and those without diabetes?

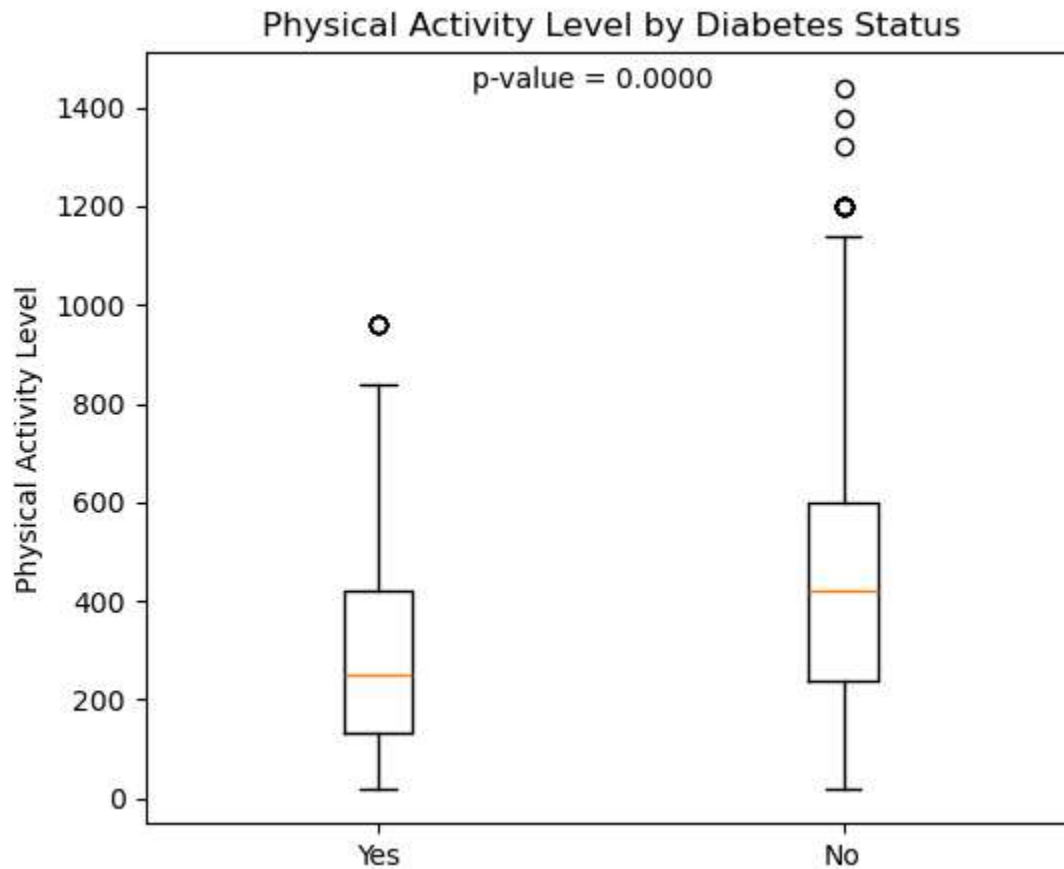
Interpretation: Since the Shapiro-Wilk p-values for both groups were less than 0.05, the data does not satisfy the normality assumption. The Q-Q plots also show clear deviations from the straight reference line, confirming that the distributions are not normally distributed.

Interpretation: Levene's test was conducted to check the assumption of equal variances between the two groups. The test produced a p-value of 0.0179, which is less than the significance level of 0.05. Therefore, the null hypothesis of equal variances is rejected, indicating that the variances between diabetic and non-diabetic groups are significantly different.

Comparing Physical Activity level Between Individuals With and Without Diabetes

A two-sample t-test was conducted to compare the mean Physical activity level between individuals with diabetes and individuals without diabetes.

Research Question: Is there a significant difference in physical activity level between people diagnosed with diabetes and those without diabetes?



Since the equal variance assumption was violated, Welch's t-test was used instead of the standard independent two-sample t-test.

Interpretation: Welch's t-test was performed to compare the mean Physical Activity Level between diabetic and non-diabetic individuals. The test produced a t-statistic of 4.8511 and a p-value of 3.29×10^{-6} , which is significantly less than 0.05.

Therefore, the null hypothesis was rejected, indicating that there is a statistically significant difference in mean Physical Activity Level between individuals with diabetes and those without diabetes.

The Welch's t-test results suggest that Physical Activity Level differs significantly between diabetic and non-diabetic individuals, indicating a possible relationship between physical activity behavior and diabetes status.

VII. Overall Conclusion

This project explored the relationship between lifestyle behaviors, physical fitness, BMI, and diabetes status using statistical analysis techniques applied to NHANES-based data.

The exploratory analysis revealed that physical activity, daily steps, and sedentary behavior are among the most influential factors related to overall physical fitness. While BMI categories showed some variation in Physical Activity Level, ANOVA and post-hoc analyses indicated that these differences were not consistently statistically significant.

The logistic regression and Welch's t-test analyses demonstrated stronger evidence of relationships between diabetes status and lifestyle behaviors, particularly Physical Activity Level.

Overall, the findings reinforce the importance of healthy activity habits and reduced sedentary behavior in promoting better health outcomes.